MSG No. 194.266 $P6_3'/mm'c$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3'm'`.

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m'`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,22,23,24]`
2	$[0,\ 0,\ -z]$	`[4,5,6,19,20,21]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8,9,16,17,18]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m'`.

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,22,23,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[4,5,6,19,20,21]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[7,8,9,16,17,18]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 7: Wyckoff site: 6g, site symmetry: .2/m'.

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,19,22]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,5,20,23]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,6,21,24]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[7,10,13,16]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[8,11,14,17]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[9,12,15,18]

Table 8: Wyckoff site: 6h, site symmetry: mm'2'

No.	position	mapping
1	$\left[x, 2x, \frac{1}{4}\right]$	[1,8,17,22]
2	$\left[-2x, -x, \frac{1}{4}\right]$	[2,9,18,23]
3	$\left[x, -x, \frac{1}{4}\right]$	[3,7,16,24]
4	$\left[-x, -2x, \frac{3}{4}\right]$	[4,11,14,19]
5	$\left[2x, x, \frac{3}{4}\right]$	[5,12,15,20]
6	$\left[-x, x, \frac{3}{4}\right]$	[6,10,13,21]

Table 9: Wyckoff site: 12i, site symmetry: .2.

No.	position	mapping
1	$[x, 0, 0]$	[1,4]
2	$[0, x, 0]$	[2,5]
3	$[-x, -x, 0]$	[3,6]
4	$\left[-x, -x, \frac{1}{2}\right]$	[7,10]
5	$\left[x, 0, \frac{1}{2}\right]$	[8,11]
6	$\left[0, x, \frac{1}{2}\right]$	[9,12]
7	$\left[x, x, \frac{1}{2}\right]$	[13,16]
8	$\left[-x, 0, \frac{1}{2}\right]$	[14,17]
9	$\left[0, -x, \frac{1}{2}\right]$	[15,18]
10	$[-x, 0, 0]$	[19,22]
11	$[0, -x, 0]$	[20,23]
12	$[x, x, 0]$	[21,24]

Table 10: Wyckoff site: 12j, site symmetry: m..

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	[1,8]
2	$\left[-y, x-y, \frac{1}{4}\right]$	[2,9]
3	$\left[-x+y, -x, \frac{1}{4}\right]$	[3,7]
4	$\left[x-y, -y, \frac{3}{4}\right]$	[4,11]
5	$\left[y, x, \frac{3}{4}\right]$	[5,12]
6	$\left[-x, -x+y, \frac{3}{4}\right]$	[6,10]
7	$\left[x-y, x, \frac{3}{4}\right]$	[13,21]

continued ...

Table 10

No.	position	mapping
8	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[14,19]`
9	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	`[15,20]`
10	$\left[x,\ x-y,\ \frac{1}{4}\right]$	`[16,24]`
11	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	`[17,22]`
12	$\left[-y,\ -x,\ \frac{1}{4}\right]$	`[18,23]`

Table 11: Wyckoff site: `12k`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,22]`
2	$[-2x,\ -x,\ z]$	`[2,23]`
3	$[x,\ -x,\ z]$	`[3,24]`
4	$[-x,\ -2x,\ -z]$	`[4,19]`
5	$[2x,\ x,\ -z]$	`[5,20]`
6	$[-x,\ x,\ -z]$	`[6,21]`
7	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[7,16]`
8	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	`[8,17]`
9	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	`[9,18]`
10	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[10,13]`
11	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	`[11,14]`
12	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	`[12,15]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$[x-y,\ -y,\ -z]$	`[4]`
5	$[y,\ x,\ -z]$	`[5]`
6	$[-x,\ -x+y,\ -z]$	`[6]`
7	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	`[9]`
10	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[12]`
13	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	`[13]`
14	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[14]`
15	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	`[16]`
17	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	`[17]`
18	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[18]`
19	$[-x,\ -y,\ -z]$	`[19]`

continued ...

Table 12

No.	position	mapping
20	$[y,\ -x+y,\ -z]$	[20]
21	$[x-y,\ x,\ -z]$	[21]
22	$[-x+y,\ y,\ z]$	[22]
23	$[-y,\ -x,\ z]$	[23]
24	$[x,\ x-y,\ z]$	[24]